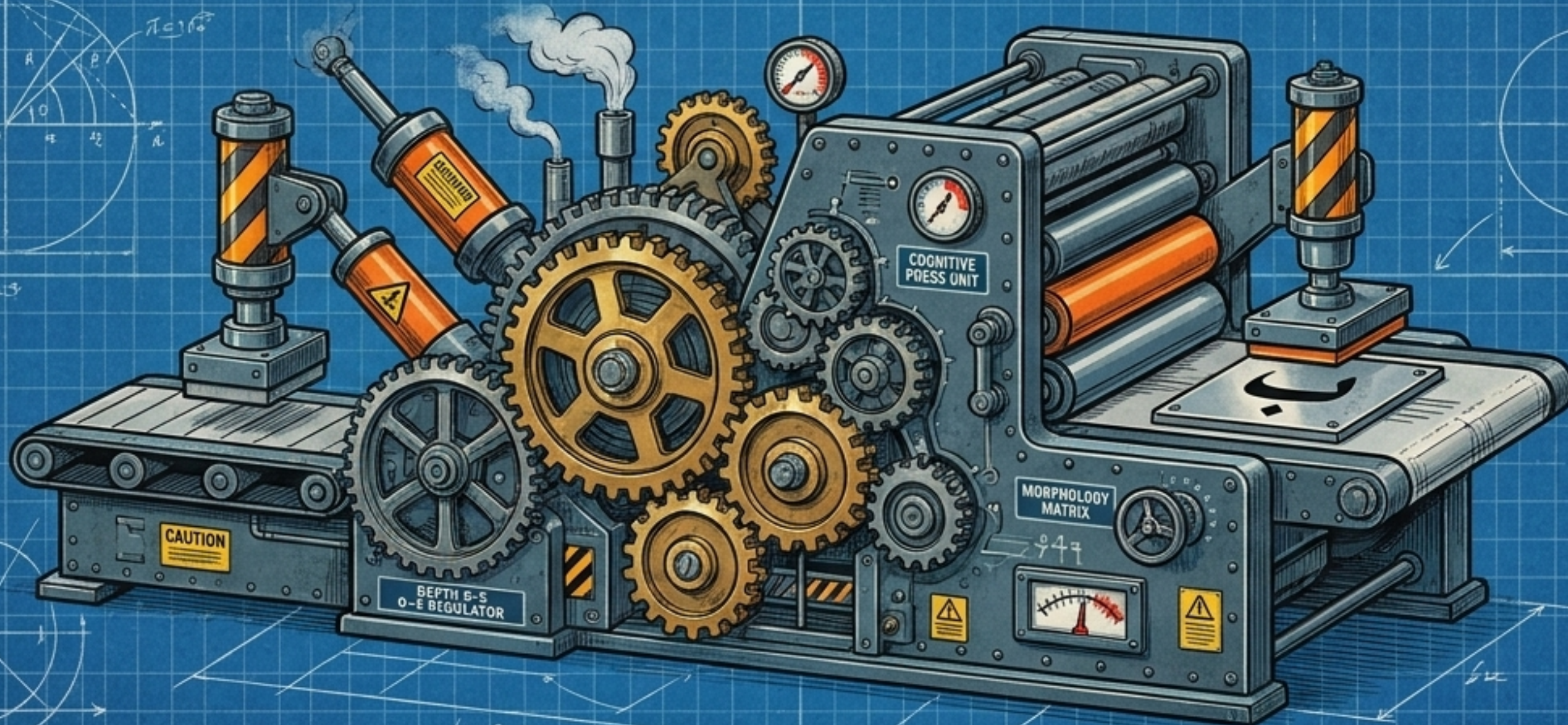


DEPTH BEFORE DERIVATION

The Operator's Manual for Orthographic and Morphological Stabilization



Not every change changes the same kind of thing.

THE FLAT MODEL OF A LAYERED HISTORY

THE PREMISE

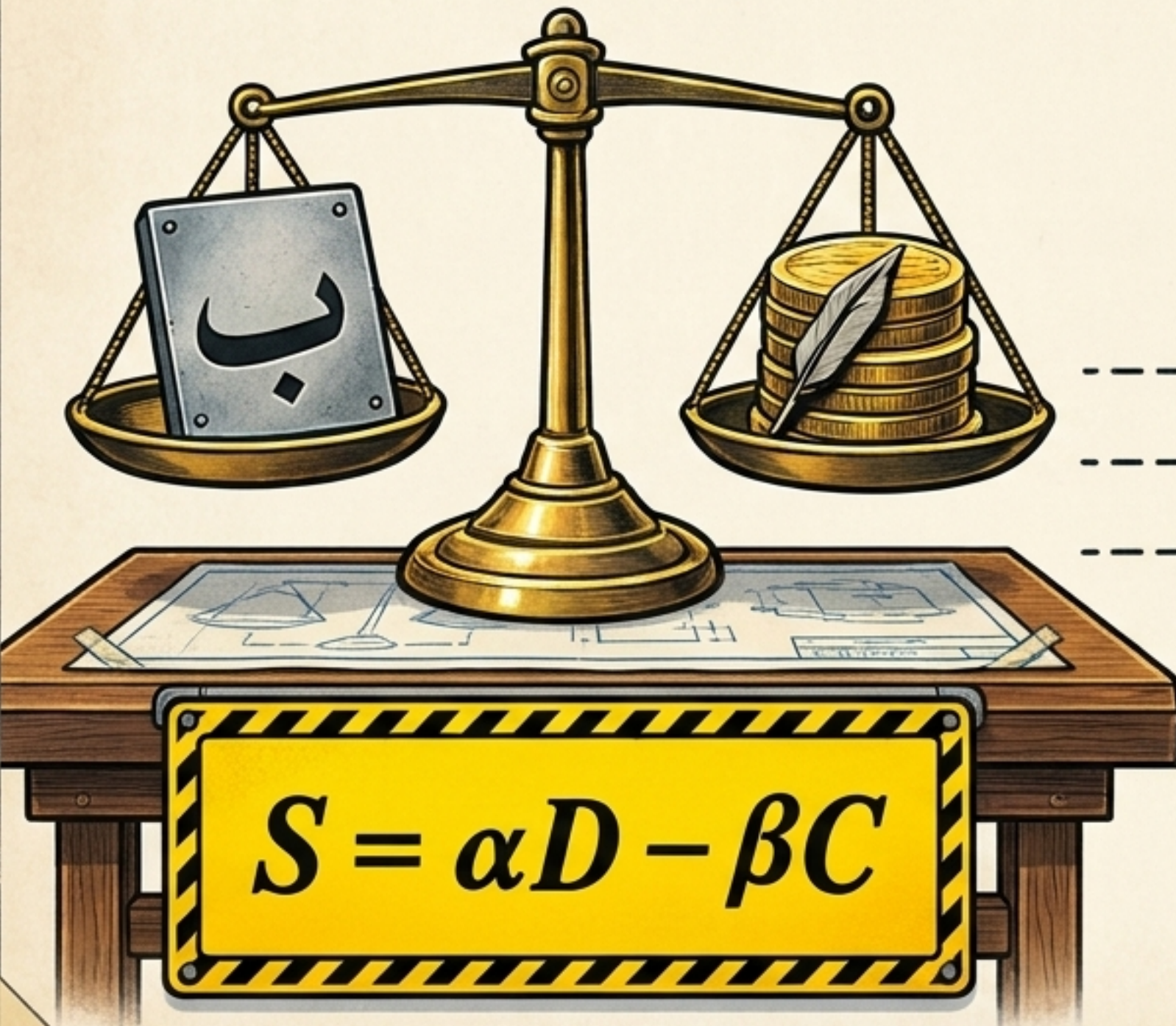
Scripts stabilize when the informational gain of a modification (D) outweighs its production cost (C).

THE SUCCESS

Perfectly models why we dot letters: low production cost traded for high phonemic distinction.

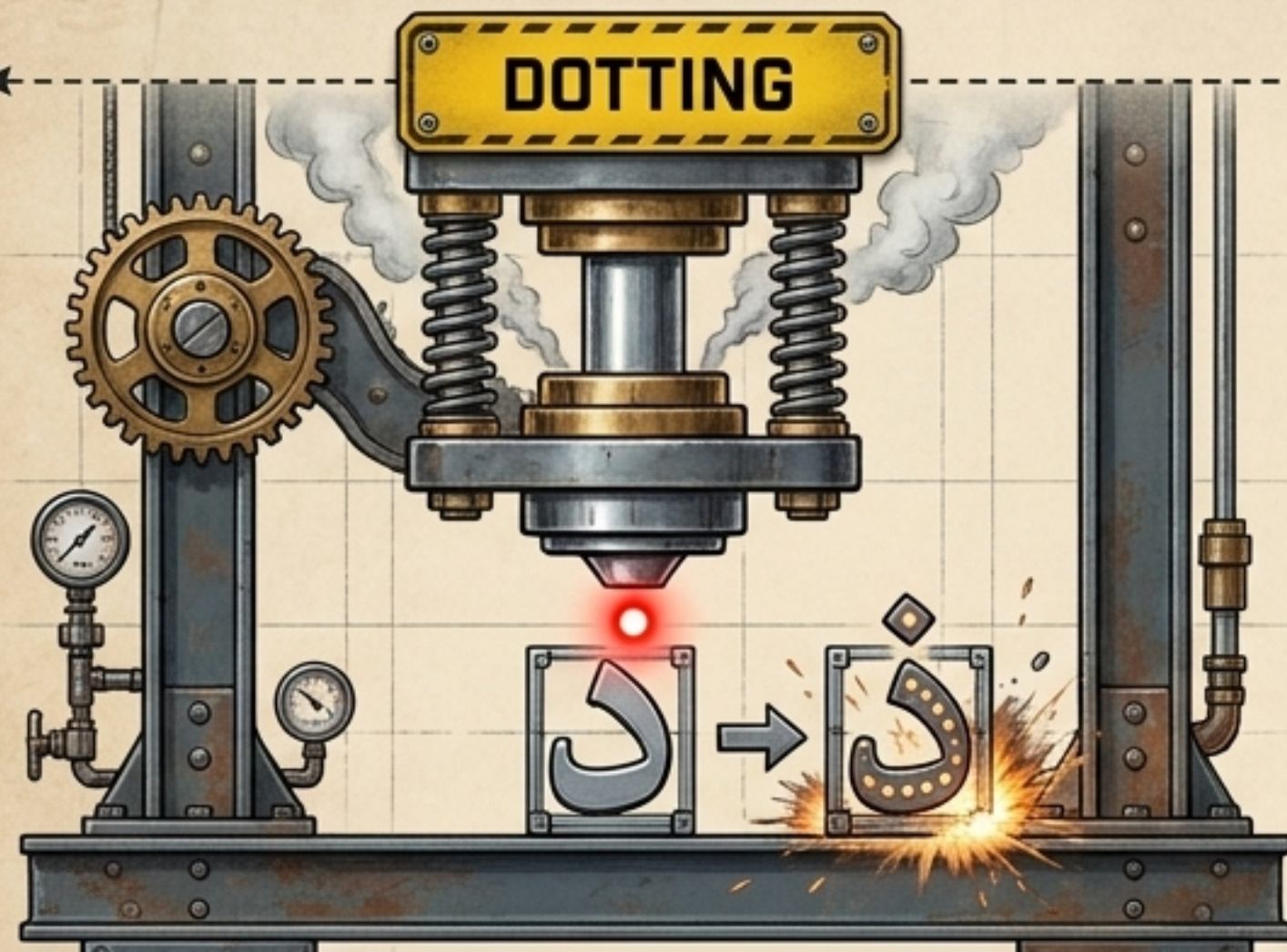
THE BLIND SPOT

Collapses distinct operations into a single trade-off, treating every stabilizing event as a change in degree, not a change in kind.


$$S = \alpha D - \beta C$$

TWO OPERATIONS OF A DIFFERENT KIND

DOTTING



Changes a grapheme. Reads the inventory and adds one new member. The generative rule remains untouched.

POSITIONAL REGULARIZATION



Changes the generative rule. Replaces an unbounded combinatorial function with a simple 4-way positional system.



**ONE WRITES INTO THE SPACE OF ADMISSIBLE FORMS.
THE OTHER REWRITES THE RULE OF ADMISSION ITSELF.**

ADDING THE DEPTH AXIS

Depth 1: Change the Generator

$M(k) : M(k-1) \rightarrow M(k-1)$.
Acts on the machinery itself.
Rewrites the rule that determines
what future forms can look like.

Depth 0: Change the Form

$M(0) : R \rightarrow R$. Acts directly
on the admissibility structure.
Executes a write operation
against a fixed rule.

THE DIAGNOSTIC MATRIX

DEPTH 0 (EXECUTE)	DEPTH 1 (CONFIGURE)
Target: The admissible set (R_t)	Target: The generative rule (F)
Notation: $R \rightarrow R$	Notation: $M \rightarrow M$
Resource Cost: C(a) paid per unit	Resource Cost: Paid once to reconstruct machinery
Orthography Example: Dotting a single letter	Orthography Example: Regulating 4-way positional forms
Morphology Example: Applying Form II to a root	Morphology Example: Fixing the 10 canonical measures
Irreversibility: Like un-stamping a page	Irreversibility: Like melting down the factory

NOTE: SELECT DEPTH BASED ON REQUIRED OPERATIONAL IMPACT.
ENSURE SAFETY PROTOCOLS FOR FACTORY RECONFIGURATION.

THE MEASURES OF f -'- l : TEMPLATES AS BOUNDED GENERATORS

The Depth 1 Configuration

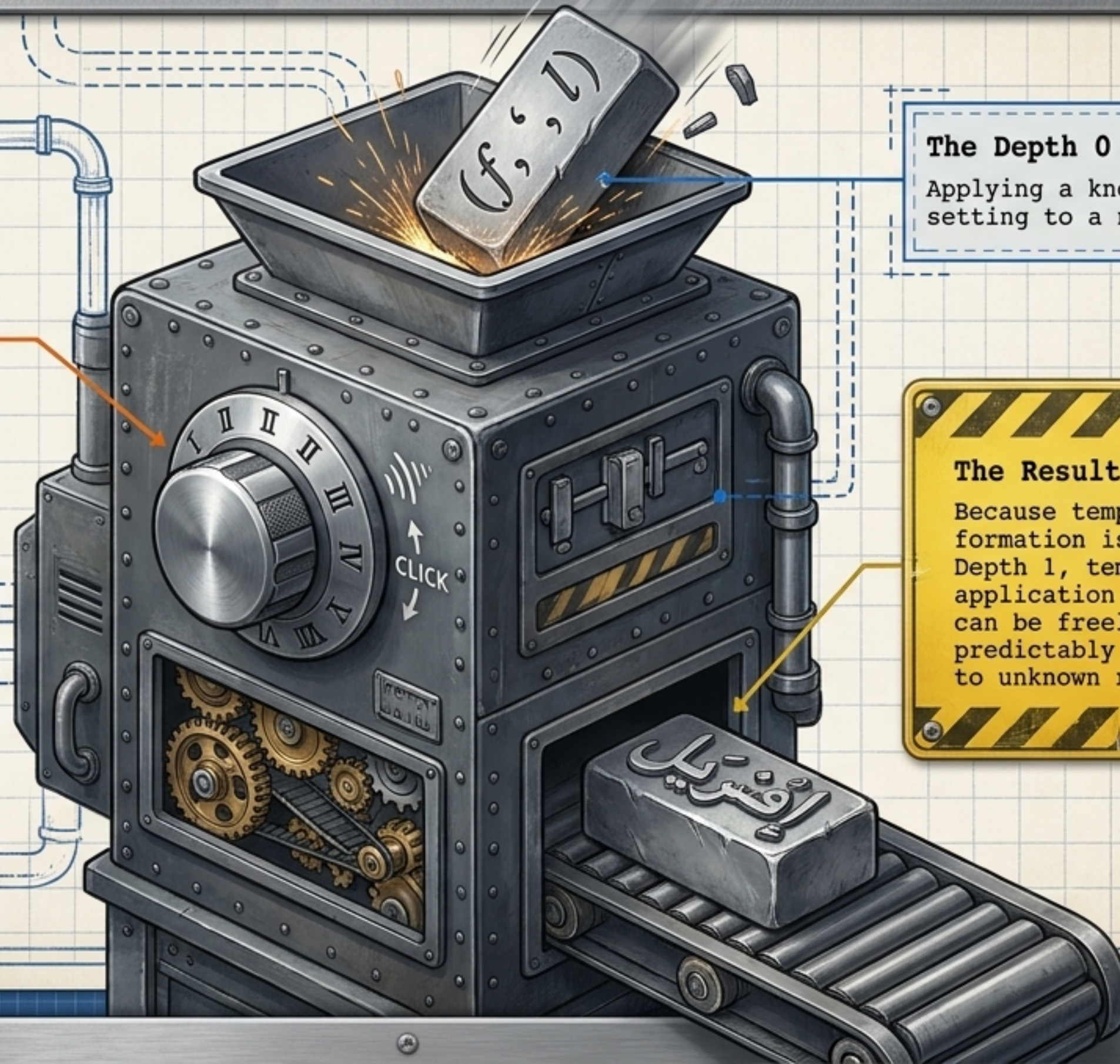
The historical stabilization of the 10 canonical measures. A closed inventory imposed on a theoretically unbounded combinatorial space.

The Depth 0 Execution

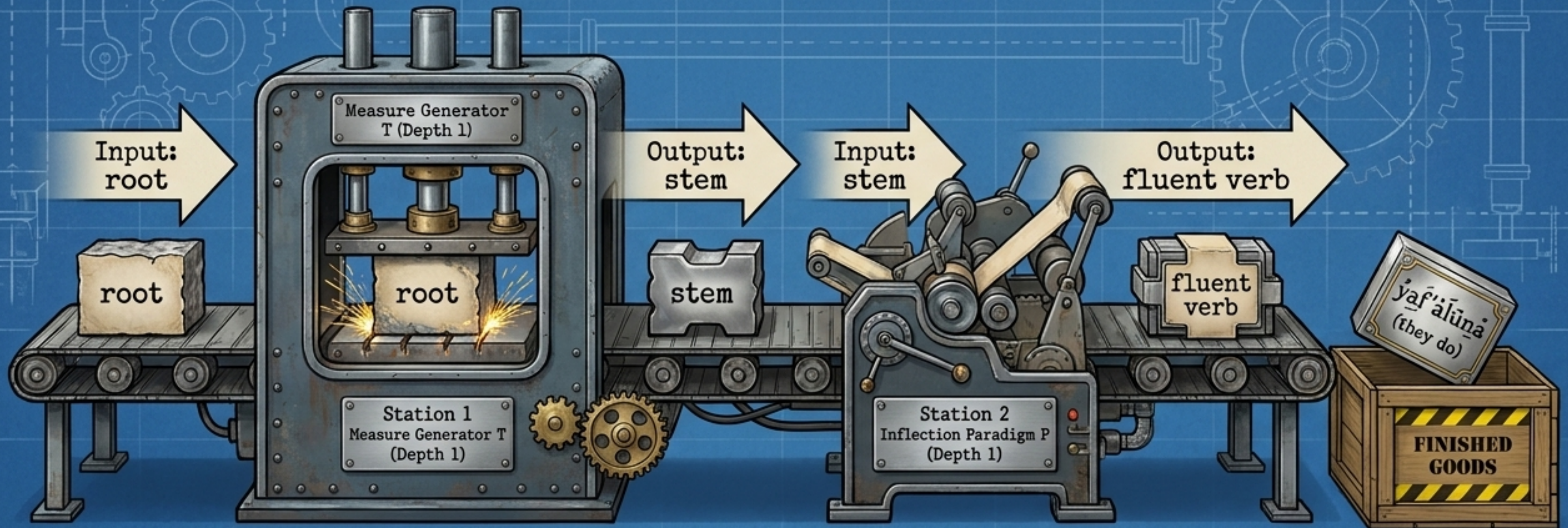
Applying a known measure setting to a new raw root.

The Result

Because template formation is locked at Depth 1, template application (Depth 0) can be freely and predictably generalized to unknown roots.



Assembly by Composition: Person, Tense, and Mood



The Mechanism

Two entirely independent fixed paradigms act in sequence. Neither modifies the machinery of the other.

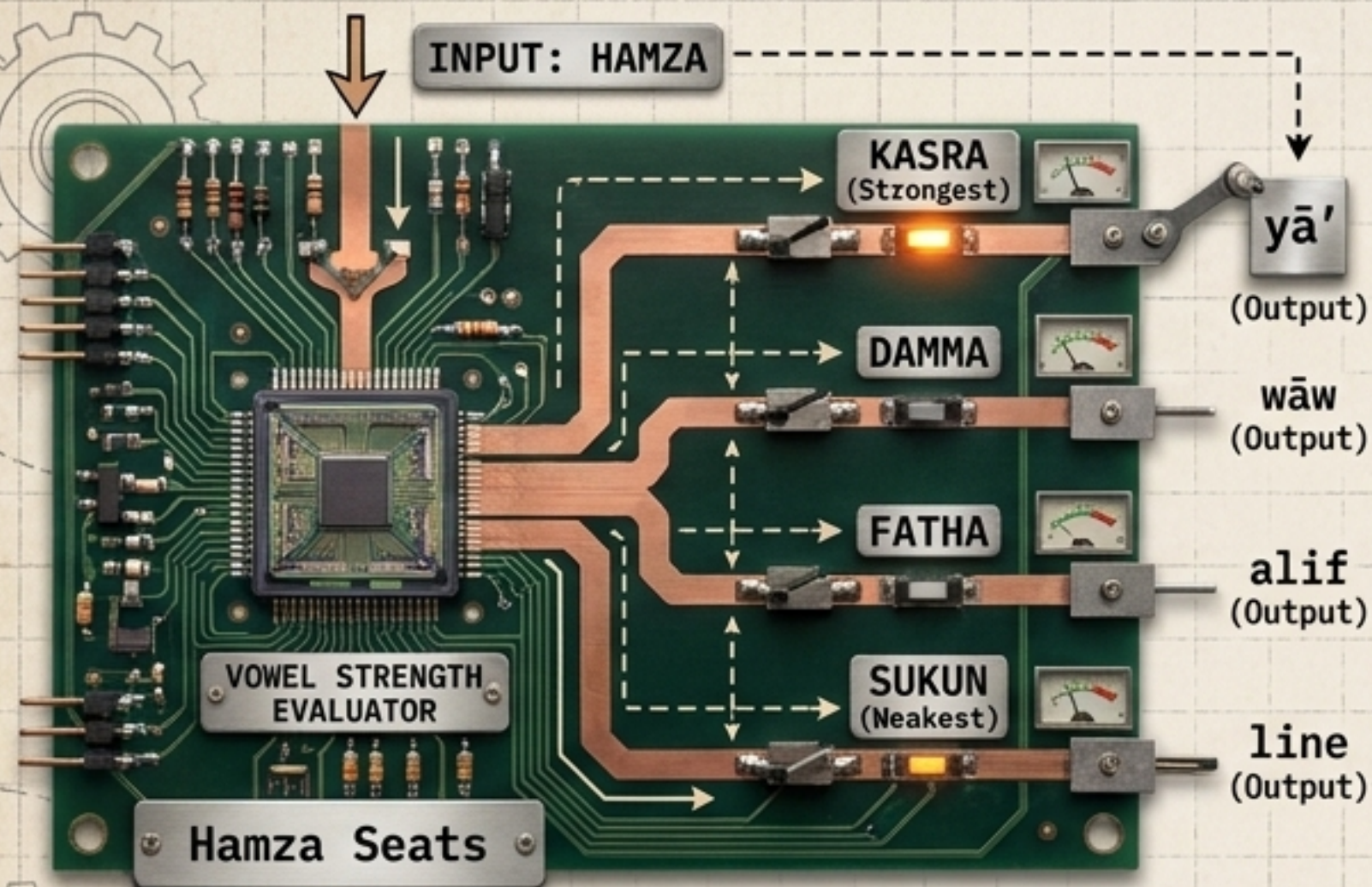
The Formula

$$\text{Stem} = T(\text{root})$$
$$\text{Word} = P(\text{stem})$$

The Insight

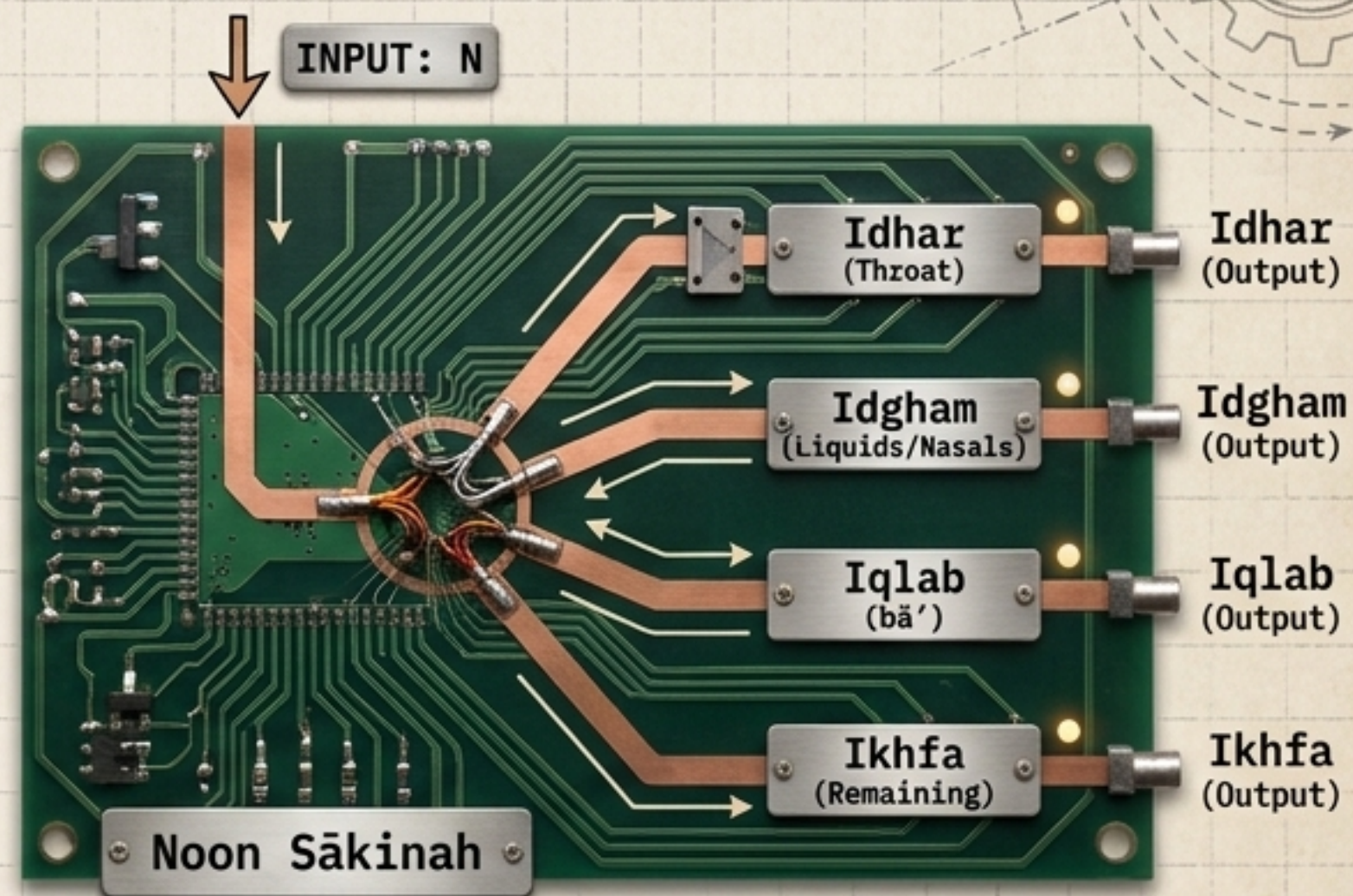
Fluent verbs are a Depth-0 composition of multiple Depth-1 paradigms. Because the machines are fixed, sequential operation is perfectly predictable.

Taxonomies as Hardwired Circuits



The Depth 1 Act

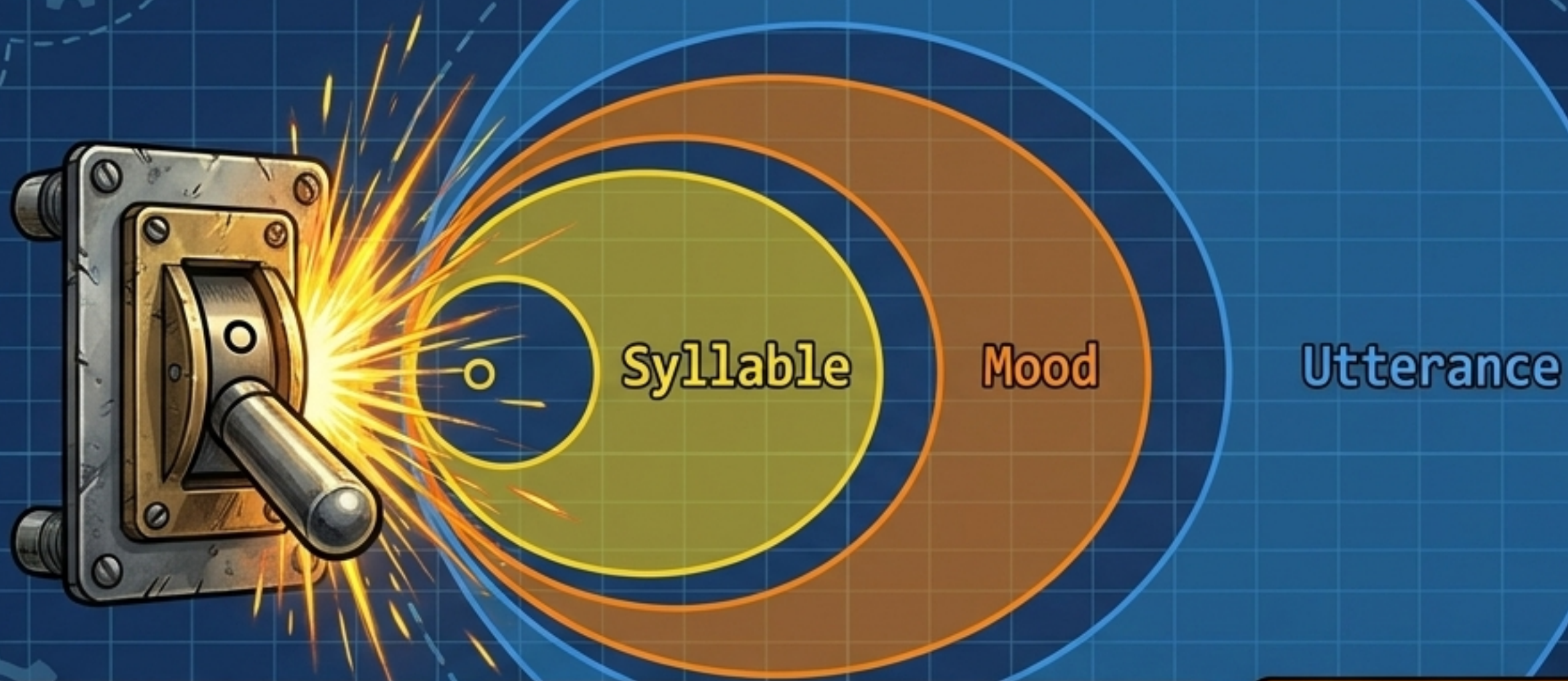
The historical codification of the ranking and the 4-way partitions. A closed taxonomy built as a hardwired circuit a speaker must memorize.



The Depth 0 Act

Routing the pulse. A single instance of Hamza finding its seat, or Noon identifying its following consonant. A single electrical pulse navigating a fixed path.

Decoupling Variables: Depth $\neq$ Scope



The Mechanism

Sukun is a null-state write—a physical switch marking the absence of a vowel.

The Proof

The identical Depth 0 operator scales massively in scope. It closes a single consonant, scales to close an entire grammatical mood (Jussive), and scales again to close a full spoken sentence (*waqf*).

Conclusion

An operation's footprint can expand across the entire linguistic system while remaining strictly Depth 0.

Decoupling Variables: Depth $\neq$ Function

The Tilde (~)

State 1

Medieval

Abbreviation
(Compression)

annu $\rightarrow$ año

State 2

Modern

Phonemic
Distinction

ano vs año



The Mechanism

The Spanish tilde began as a weak Depth-0 shorthand purely to save ink.

The Evolution

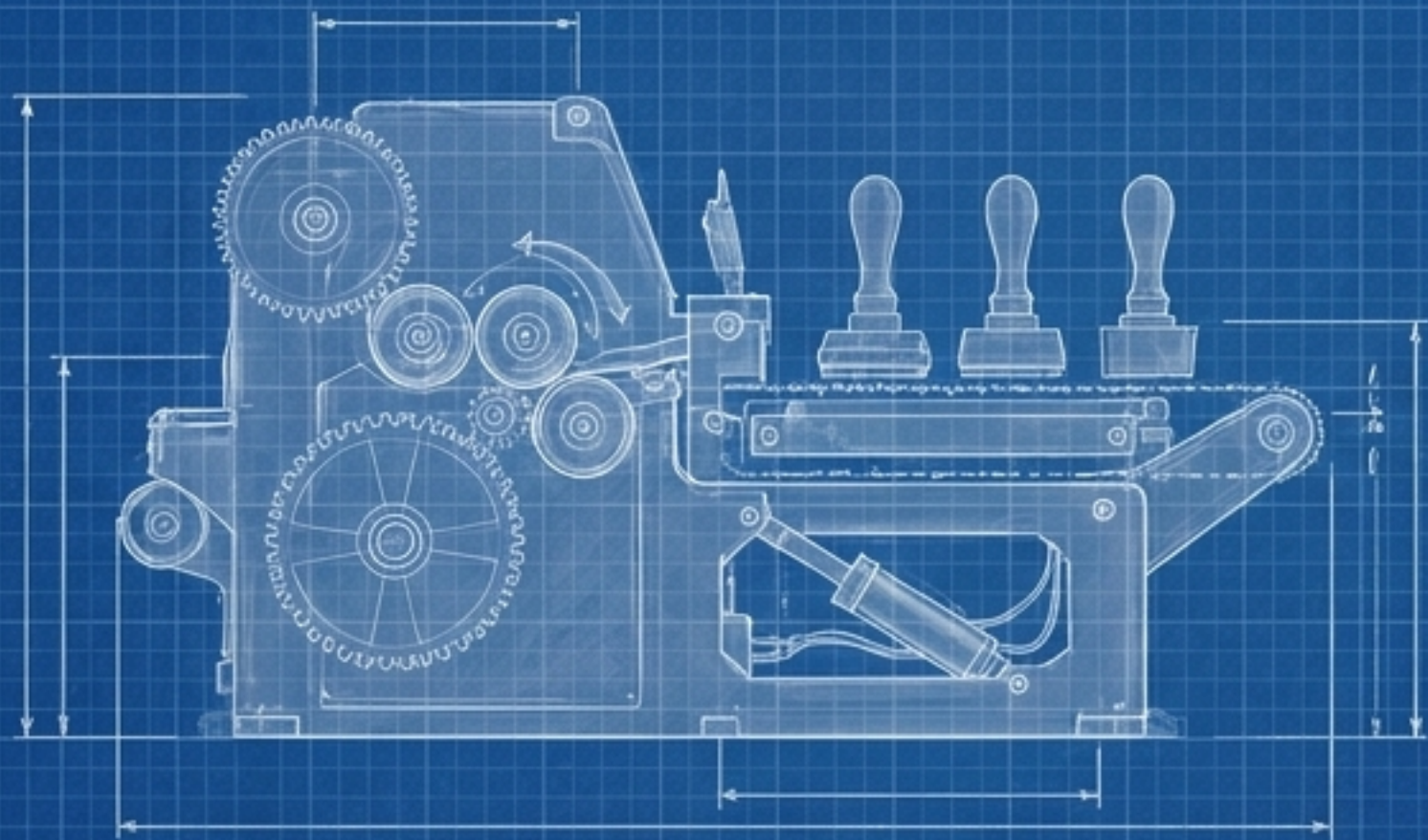
Following palatalization, it migrated to function exactly like Arabic dotting—a phonemic distinguisher.

The Proof

A mark can migrate from mere compression to core differentiation without ever leaving Depth 0.

Fixed Base: The Depth 0 Chassis never changes.

The Honest Exception: Broken Plurals



The Machine:
Computed Derivation



The Archive:
Memorized Lookup

The Anomaly

Not all morphology is a clean machine.
Broken plurals use roughly 30
patterns, only partially predictable
by phonological shape.

The Diagnosis

Defies clean categorization.
It is not an executable rule (Depth 0)
nor a modification of a rule (Depth 1).

The Mechanism

It is an interrogative lookup.
Memorized association masquerading
as computed derivation.
A messy archive, not a factory.

The Cost of Irreversibility

Depth 0 Reversal

Pays $C(a)$ for a single grapheme. Smoothly reversible if weights shift.

Depth 1 Reversal

Requires re-deriving the entire combinatorial space. You aren't restoring a lost state; you are paying to restore a lost machine.

You aren't restoring a lost state; you are paying to restore a lost machine.

$$S = \alpha D - \beta C$$

The Core Insight: The flat model treats all reversals as symmetric moves along the Cost/Distinctness line. The **Depth** axis proves they are **not**.

To understand how a system evolves—and what it cannot undo—you must track not just what changed, but what tier of the machine was rewritten.